

# Random Samples and Simple Probability Models

## Answer Key

Grade 7

### Quick Answers

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| <p>1. <math>1 \cdot \frac{1}{4}</math><br/>                 2. <math>\frac{3}{8}</math><br/>                 3. 216<br/>                 4. <math>\frac{7}{12}</math><br/>                 5. (b) the estimate this sample would give = 704, —<br/>                 6. (a) an even number = <math>1 \cdot \frac{1}{2}</math>, (b) a number greater than 7 = <math>\frac{3}{10}</math><br/>                 7. 150</p> | <p>8. (b) probability one named member is chosen = <math>1 \cdot \frac{1}{10}</math>, —<br/>                 9. (a) probability of red = <math>1 \cdot \frac{1}{5}</math>, (b) probability of red or blue = <math>1 \cdot \frac{1}{2}</math>, (c) expected number of reds in 200 draws = 40<br/>                 10. (a) fraction of the sample that was tagged = <math>\frac{3}{20}</math>, (b) estimated number of fish in the lake = 2000, —</p> |
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### What is verified

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7 of 10 problems fully machine-checked against SymPy.

— marks an answer only you can judge — problems 5, 8, 10. The worked solution states what a correct response must contain.

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### Curriculum

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**Level:** Grade 7

7.SP.A – *problems 3, 5, 7, 8, 10*

7.SP.C.5 – *problems 1, 8*

7.SP.C.7 – *problems 2, 4, 6, 9*

*Difficulty 1–5 of 5 across 17 tagged checks.*

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- 1.** Bag of 12 counters, 3 red. Probability of drawing red.

**Solution:** Every counter is equally likely, so the probability is favourable over total:  $\frac{3}{12}$ , which reduces to  $\frac{1}{4}$ .  $\frac{1}{4}$

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- 2.** Spinner with 8 equal sections, 3 shaded.

**Solution:** The sections are equal, so they are equally likely outcomes:  $\frac{3}{8}$ . It is already in lowest terms.  $\frac{3}{8}$

3. Random sample of 50 students, 18 walk; school of 600.

**Solution:** The sample says a fraction  $\frac{18}{50}$  of students walk. Applying that fraction to the whole school:  $\frac{18}{50} \times 600 = 0.36 \times 600 = 216$  students. Because the sample was random, it is fair to scale it up like this. 216

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4. Box of 12 counters, 5 yellow. Probability of not yellow.

**Solution:** If 5 of the 12 are yellow then  $12 - 5 = 7$  are not, so the probability is  $\frac{7}{12}$ . (Equivalently  $1 - \frac{5}{12} = \frac{7}{12}$ .)  $\frac{7}{12}$

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5. School of 800; the 25 basketball-team members surveyed, 22 choose basketball.

**Solution:** (a) **Instructor-judged.** No. The sample is not random and not representative: only members of the basketball team were asked, and they are far more likely to name basketball than a randomly chosen student would be. Full credit needs that reason — the sample was drawn from a group already tied to the answer — and not simply the word “biased”.

(b) Scaling the sample anyway gives  $\frac{22}{25} \times 800 = 0.88 \times 800 = 704$  students. That is the point of the problem: the arithmetic is correct and the estimate is still worthless, because the sample was not random. 704

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6. Spinner numbered 1 to 10.

**Solution:** (a) The even numbers are 2, 4, 6, 8, 10 — five of the ten outcomes:  $\frac{5}{10} = \frac{1}{2}$ .  $\frac{1}{2}$

(b) Greater than 7 means 8, 9 or 10 — three outcomes, and 7 itself does not count:  $\frac{3}{10}$ .  $\frac{3}{10}$

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7. Random sample of 80 bulbs, 6 faulty; batch of 2000.

**Solution:** The faulty fraction in the sample is  $\frac{6}{80} = 0.075$ , so the batch is estimated to hold  $0.075 \times 2000 = 150$  faulty bulbs. 150

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8. Club of 200 members; two ways of choosing 20.

**Solution:** (a) **Instructor-judged.** Method 2. Numbering all 200 members and drawing 20 at random gives every member the same chance of being picked. Method 1 reaches only the members who arrive early, and those members may differ from the rest in ways that matter. Full credit needs the equal-chance idea stated about the whole membership.

(b) Twenty of the 200 members are chosen, and each is equally likely, so a named member has probability  $\frac{20}{200} = \frac{1}{10}$ .  $\frac{1}{10}$

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**9.** Jar of 20 marbles: 4 red, 6 blue, 10 white, drawn with replacement.

**Solution:** (a)  $\frac{4}{20} = \frac{1}{5}$ .  $\frac{1}{5}$

(b) Red or blue is  $4 + 6 = 10$  of the 20 marbles:  $\frac{10}{20} = \frac{1}{2}$ .  $\frac{1}{2}$

(c) A probability of  $\frac{1}{5}$  over 200 draws gives an expected  $\frac{4}{20} \times 200 = 40$  reds. This is a long-run expectation, not a promise: 37 or 44 reds would be perfectly ordinary. 40

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**10.** 300 fish tagged; a later random sample of 60 contains 9 tagged.

**Solution:** (a)  $\frac{9}{60} = \frac{3}{20}$  of the sample was tagged.  $\frac{3}{20}$

(b) If the sample is representative, the same fraction of the whole lake is tagged. With  $x$  fish in the lake,  $\frac{300}{x} = \frac{9}{60}$ , so  $9x = 300 \times 60 = 18000$  and  $x = 2000$  fish.  $x = 2000$

(c) **Instructor-judged.** A correct response names one assumption — that the tagged fish mixed evenly back through the lake, that every fish was equally likely to be caught the second time, or that no fish entered or left in between — and then gives the direction of the error: if tagged fish are easier to catch, the sample holds more than its fair share of them, the tagged fraction comes out too large, and the estimated lake population comes out too small. Full credit needs an assumption *and* the direction of the error.

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